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The importance of organization type: Construction sector perceptions of low-carbon policies and measures

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ABSTRACT

To facilitate decarbonization in the EU's building sector, the regulatory focus is shifting towards a life cycle perspective of emissions. The EU's path in tackling embodied emissions is under development in the revision of the Energy Performance of Buildings Directive, which is utilizing Level(s) framework and EN 15879 as a methodological foundation. Member States are at different stages in their building life cycle regulation plan and Finland is preparing normative regulations to limit the carbon footprints of new buildings. We conducted a survey on how the different actors in the construction sector perceive new low-carbon policies under development in Finland. Our findings indicate that carbon footprint policies were generally seen as functional. However, perceptions differed significantly depending on the respondent's organization type. The respondents involved in the construction design stage generally seemed more positive about the proposed regulations, whereas advocates (associations and interest groups) were more cautious than the organizations they represented. These findings help clarify the implementation and development needs of low-carbon policies.

1. Introduction

1.1. Background

In 2020, the building sector was responsible for 36 per cent of the world's final energy use and 37 per cent of its energy-related CO_2 emissions (United Nations Environment Programme, 2021).[1] The sector is also responsible for half of all material extraction, one-third of waste generation, and one third of water consumption in the EU (European Commission, 2021). Numerous policies and measures to curb the environmental impacts of the building sector have been implemented or are under preparation. In this paper, we study the construction sector actors' perceptions of the new low-carbon policies and examine how these perceptions differ according to organization type.

The EU aims to almost decarbonize the entire building stock by 2050. All new buildings in the EU have had to meet the nearly Zero Energy Buildings (nZEBs) standards since 2021.[2] However, the actual implementation of these policies is left to the individual Member States, and local policies are non-uniform and difficult to compare (Garcia and Kranzl, 2018; D'Agostino and Mazzarella, 2019). To facilitate the energy-efficiency transition of existing building stock, the Member States must craft long-term renovation strategies, to forge the path to decarbonized EU building stock by 2050 (European Union, 2018).

Ensuring the high-level energy efficiency of new buildings and renovating old buildings is not enough to fully tackle the building sector's greenhouse gas (GHG) emissions. Producing building materials and carrying out construction activities causes a substantial amount of *embodied emissions*. These, together with demolition activities, represent a large and growing portion of the sector's environmental impact. For modern, energy-efficient new buildings, the share of GHG emissions arising from construction materials and processes may be 50 per cent or more of total life cycle emissions (e.g., Azari and Abbasabadi, 2018; Koezjakov et al., 2018; Röck et al., 2020). Reflecting this, the focus on curbing building sector emissions is now moving towards the life cycle

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[1] The relative energy use and emissions contribution remained fairly flat year-on-year (In 2019 35% and 38% respectively), with lower levels of activity in both construction and total numbers due to the COVID-19 pandemic.

[2] nZEBs have very high energy efficiency and to a significant extent provide the required small amount of energy from renewable sources (European Union, 2010).

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climate impacts of buildings, covering all the above-mentioned emission phases, both at the EU level and within many EU Member States.

The EU's path for tackling buildings embodied emissions is dependent on the revision of the Energy Performance of Buildings Directive (EPBD), the methodological development of which calls on the Level(s) framework. Level(s) was launched in 2020 (European Commission, 2020), and its methodology is based on existing European standards for buildings life cycle assessment (most relevantly EN 15978, EN 15804, EN 15643–5). Parallel to the EPBD revision, various EU Member States are currently creating their own national methods for assessing the impacts of buildings' life cycles. Denmark, the Netherlands, and France limit the life cycle carbon emissions for a large share of new buildings, whereas Germany and Switzerland have life cycle carbon requirements for public buildings. Sweden[3] and Norway now require Climate Declaration for new buildings. Both Finland and Sweden are currently in the process of introducing life cycle carbon regulation: Finland aims to introduce CO_2 limits for a large proportion of new buildings by 2025, Sweden by 2027 (BPIE, 2021a).

Finland is preparing a quantifiable method for assessing buildings lifetime carbon based on Level(s) and European standards (Kuittinen, 2019). The method has been in the trial phase since 2019 and has undergone several modifications based on piloting experiences and feedback. The system boundary at the building level is based on life cycle modules of EN 15978; emissions related to construction products and worksites in production and construction process stage, the use and maintenance of a building in the use-stage, as well as its demolition and the recycling of the materials in the end-of-life stage. The assessed building parts include the energy consumed, ground and reinforcement elements and structures on the building cite. For load-bearing and supplementary structures of the building only minor auxiliary items that do not come with the products are excluded. Assessed building systems include heating and cooling, air conditioning, water, electrical, sprinklers and lifts (Kuittinen, 2019). The emission values for building parts will be obtainable from a free-of-charge national database CO2data or from an Environmental Product Declaration when such is available.

A carbon footprint limit value is planned as a measure to cap the life cycle emissions of most types of new buildings in Finland. Finland has collaborated actively with other Nordic countries in developing both the assessment method and the building product emission database (Kuittinen and Häkkinen, 2020).[4] Coordinated development of national building life cycle regulation strategies may be highly beneficial at the wider EU level. If successful, early actions taken by the forerunners will demonstrate that the building carbon footprint regulation can be implemented and accepted by stakeholders.

1.2. Policy acceptance and implementation

For the building carbon life cycle regulation to succeed, stakeholder perceptions and attitudes are crucial, as recognized in the BPIE (2021b) summary report. Evidence shows that stakeholder engagement has impacts on regulation development (Bijlsma et al., 2011). Rasmussen and Toshkov (2013) observed that stakeholder consultations led to a slower legislative process in the EU, but whether this outweighs the potential gains from consultations remains unclear. In the construction sector, Habert et al. (2020) emphasize the role of the stakeholder groups in the development and acceptance of the building carbon footprint regulation. They note that the professional organizations of, for example, architects, engineers, real estate companies, financers, and property owners can help establish good practices.

Even a single construction project involves many different actors, and possible negative attitudes may obstruct a project's implementation (Olander and Landin, 2005). Previous construction sector research has found differences in organizational cultures (Ankrah and Langford, 2005) and in the perceptions of, for example, safety risks (Jin et al., 2017; Zhao et al., 2016), project performance indicators (Lai and Lam, 2010), and effective management (Fraser and Zarkada-Fraser, 2003).

A study by Serpell et al. (2013) revealed that construction companies' measures to support sustainability depend greatly on the company's size and core business. The results show that larger companies and companies involved in infrastructure construction implement sustainable construction actions more frequently than smaller companies or companies involved in building construction. This is the consequence of greater client demand for sustainable projects and greater awareness of the environmental protection of large firms and infrastructure companies. Kangas et al. (2018) found that from the viewpoint of energy service companies whose core business was energy efficiency improvements, builders, developers, and building managers, as well as many public bodies, were not interested in sustainable building practices.

In this study, the focus was on stakeholder attitudes towards the proposed new legislation. To the best of our knowledge, the differences in construction sector actors' perceptions of carbon footprint policies have not been studied before. In addition to overall acceptance of a policy reform, we need to account for possible heterogeneity in the views of construction sector actors. On this basis, we conducted a survey to determine the views of construction sector actors on regulating buildings' carbon footprints. We surveyed the acceptance of the carbon footprint regulation, the limit value for the footprint, and the most critical aspects on which the regulation may have conflicting impacts. To gain a deeper understanding of our findings, we studied the differences between the views of the construction sector respondents.

We need to better understand how different stakeholder perceptions can influence the development and acceptance of policy development (Balane et al., 2020), and how views are established along this process. Investigating the varying perceptions of different organization types should identify the existing positions, knowledge gaps, and areas that need development. Our study makes a significant empirical contribution to understanding organizational differences and the general acceptance of emerging carbon footprint regulation. It is extremely timely, as frontrunners can influence building carbon footprint policies in other EU countries and beyond.

The article proceeds with the survey details and the methods employed in Section 2. Section 3.1 presents the general results of the survey, and Section 3.2 delves into organizational differences. Finally, in Section 4 we draw our conclusions.

2. Material and methods

2.1. Survey design and respondents

The survey was originally carried out as part of a climate impact assessment of the Finnish Land Use and Building Act reform in collaboration with the Finnish Ministry of the Environment (MoE), and it focused on the carbon footprint policies related to the construction sector. The survey was conducted using Webropol in November 2020. The version of the footprint calculation method available at the time of the survey was based on the report by Kuittinen (2019).

We carefully designed the survey to cover all the relevant aspects of the planned carbon footprint measures and their potential impacts, as well as the respondents' background information. This was ensured by piloting the survey and collecting feedback from carbon construction experts. The survey asked i) general questions on how the respondents weighed low-carbon benefits against other building practice objectives, and ii) questions that were directly linked to the planned Finnish carbon footprint policy measures. The survey utilized Likert-scale claims, nominal questions, and open-ended questions. Due to the novelty of the planned policies, the respondents were given some background

[3] See von Malmborg et al. (2023) for statement analysis on the Climate Declaration implementation in Sweden

[4] The development of the Finnish building life cycle assessment method is discussed in detail in Kuittinen and Häkkinen (2020).

information on the policy measures (see Table 1).

The target group of the survey were different construction sector actors such as designers, construction companies, construction material producers, interest groups, and researchers (see Appendix A for how the recipients were found). The survey invitation was sent to 301 recipients, and included both a personal link to respond to the survey and a public link that could be shared in, for example, the recipients' organizations. Overall, the survey received 86 responses.

Table 2 presents the descriptive statistics of the respondents. The results imply wide representation of different construction sector actors. The majority of the organizations operating at the regional level were based in Uusimaa, the most densely populated region in Finland.

2.2. Data analysis methods

The survey analysis started with an exploratory data analysis of construction sector views of the emerging low-carbon policies under study. This involved graphical analysis and comparison of means stratified with the variables depicted in Table 2 and the construction sectors in which the respondents' organizations operated. The emerging hypothesis, based on the examination and the literature review, was that views on low-carbon policies are likely to differ according to organization and respondent characteristics.

The organization type variable shown in Table 2 is further categorized into design stage and construction stage organizations. Examining them together is in line with earlier research, which has identified and grouped different stakeholders. Examples of this are the comparison of designers and builders in Zhao et al. (2016) and the classification of the energy-related stakeholders of buildings' life cycles into the design-, construction- and operation stages in Xu et al. (2019).

Table 1
Background information on planned policy measures given to survey respondents.

Measure	Background information
The carbon footprint assessment method	A carbon footprint assessment method is used to calculate buildings' impacts on the climate. The assessment method covers a building's life cycle, from construction material production through transport and construction site activities, use and maintenance, to demolition and recycling. The method can be used to assess the carbon footprint (life cycle GHG emissions) and carbon handprint (climate benefits of a building), and is prepared by the MoE.
Climate Declaration	The Climate Declaration would report the life cycle emissions of the building (carbon footprint + carbon handprint) that have been calculated by using the carbon footprint method provided by the MoE. The Climate Declaration would have to be provided for new housing and service buildings. It would also have to be provided in the case of large-scale renovations. Exceptions would be detached houses. A Climate Declaration would have to be provided when applying for a construction permit.
Carbon footprint limit value	A carbon footprint limit value is a cap that the carbon footprint of buildings cannot exceed. The carbon footprint limit value would not apply to detached buildings and for those houses that do not need to meet EU's nearly zero energy building requirements. The limit value would mean changes to current building practices. The main construction material and the main source of heating have the biggest impact on the carbon footprint of a building. According to a preliminary assessment*, a 10–20% lower limit value would entail choosing either a low-carbon heating system or low-carbon building material, and a 30–40% lower limit value would necessitate low carbon solutions in both.

* The background information provided to respondents was based on ongoing unpublished assessment at the time. A Ministry of the Environment (2021) commissioned report has been published post-survey.

Table 2
Descriptive statistics of respondents.

Characteristics	N	Characteristics	N
Sample size	86	**Organization size:**	
Area of operation:		less than 10 employees	10
National	47	10–50 employees	20
Regional	39	51–250 employees	25
Working for a firm:		over 250 employees	30
Yes	59	**Organization type:**	
No	27	Construction product industry	21
Professional status:		Association or advocacy	17
Management	28	Consulting company	11
Middle management	12	Construction company or turnkey home supplier	10
Specialist	35	Architecture or other design office	10
Staff member	4	Research institute or university	7
Teaching or research	4	Real estate company or rental apartment owner	5
Entrepreneur	2	Building supervision or other authority	3
Other	1	Other	2

Confirmatory data analysis focused on Likert-scale claims regarding the determinants of the acceptance of the carbon footprint assessment method and the effectiveness of the carbon footprint limit value. Examining them both enhanced the robustness of the results, as it can catch the critique aimed explicitly at the Finnish method. To uncover the drivers of the views on these variables, we conducted discrete choice regression analyses (see, e.g., Train, 2009). The five Likert-style scales were first transformed into binary variables such as agreeing with the statements (4 or 5 on the Likert scale). These binary views were then analysed using a Binomial Probit model, which is derived from a random utility framework. The derived utility consists of observable and unobservable parts, with the random component assumed to be normally distributed. It is estimated using the maximum likelihood method (see e.g., Train, 2009 or Greene, 2012 for details).

Based on the binomial regression discovery, which pointed to organization type being the main driver of the differences, we proceeded to perform pairwise comparisons of the statistical difference between the stakeholder response groups, using the non-parametric Mann–Whitney U-test. The U-test assumes independent samples and does not require normally distributed differences.

3. Results and discussion

3.1. Perceptions of carbon footprint policies

3.1.1. The carbon footprint assessment method

The carbon footprint assessment method (see Table 1 for description) was well known by the respondents. Most indicated either having used the method or having familiarized themselves with it by reading about it. Only five per cent had not heard of it before. However, in the open-ended questions, some respondents wished the method had clear guidelines and routines, and that it was simple, and easy to use.

Almost 60 per cent of the respondents agreed that the carbon footprint assessment method was an appropriate tool for assessing the climate impacts of construction (see Fig. 1). This finding is consistent with that of Díaz-López et al. (2021), who found that in Spain, experts believed that the strengths and opportunities of the Level(s)-based regulation of building performance outweighed its weaknesses and threats. However, almost a third of the respondents were critical of the method. In the open-ended questions they raised concerns about the carbon footprint calculation being too laborious for detached houses. They also raised concerns about the validity of the method (e.g., whether it includes all emissions and how accurate it is).

3.1.2. Climate declaration

When applying for a building permit, the carbon footprint calculated

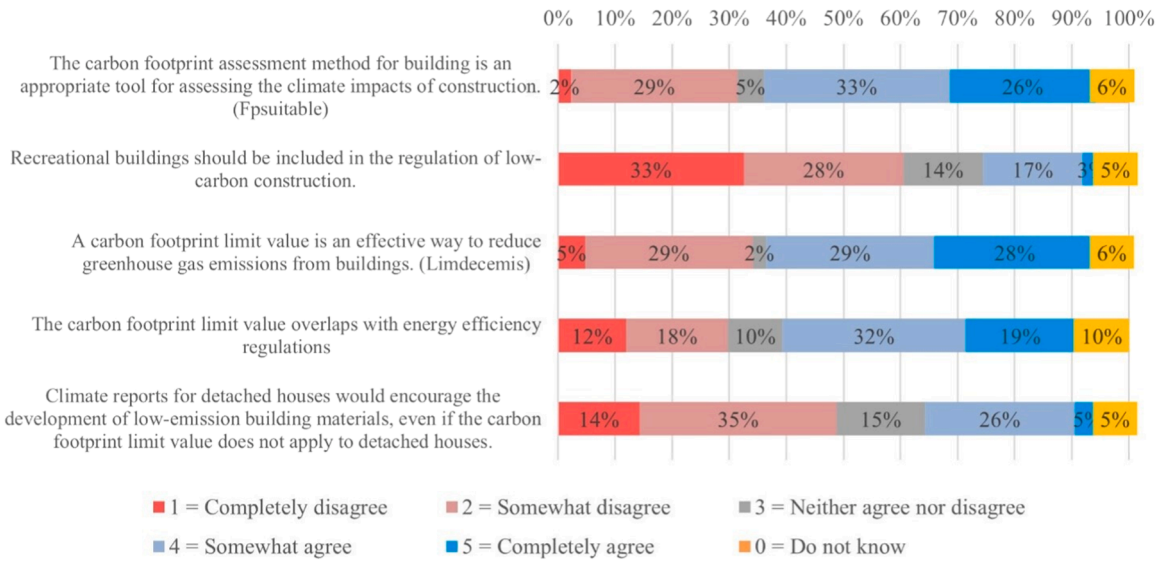


Fig. 1. Appropriateness of carbon footprint value policy.

using the carbon footprint assessment method must be declared via the Climate Declaration (see Table 1 for a more detailed description). Fig. 2 shows a list of claims pertaining to the Climate Declaration. The respondents' views on Climate Declaration were mostly positive. Over half of them agreed that preparing a Climate Declaration would both encourage the reduction of emissions in large-scale renovation projects and provide an opportunity to promote low-carbon construction through urban planning. Most did not see that the amount or quality of renovations would be negatively affected by a climate reporting mandate. A third of the respondents, however, were concerned that preparing a Climate Declaration would incur significant additional construction costs. Still, in their responses to open-ended question, some respondents argued that the additional costs would decrease as the carbon footprint assessment becomes part of the normal design process. The main concern about Climate Declarations was related to the role of building supervision. Only a quarter of the respondents believed that municipalities' building supervision would be able to monitor Climate Declarations.

3.1.3. Carbon footprint limit value

Almost 60 per cent of the respondents agreed that a carbon footprint limit value (see Table 1 for description) would be an effective way to reduce buildings' GHG emissions (see Fig. 1). About a third of the respondents did not see carbon footprint limit values as an effective tool, and in the open-ended questions they raised concerns about the difficulty of setting the value and a lack of ambition for it. They also feared that it would overlap with the energy efficiency regulation.

On the other hand, the respondents saw no need to set a limit value for life cycle emissions for detached houses; almost 60 per cent felt that Climate Declarations for detached houses would encourage the development of low-emission building materials, even if the carbon footprint limit value did not apply to them. A majority of the respondents also felt that detached houses should be exempted from climate reporting, and recreational buildings should be exempted from low-carbon regulations overall. These views seem to suggest that actors in the Finnish construction sector prefer non-binding policy tools and self-regulation for detached houses and recreational buildings.

The survey also examined how much lower the carbon footprint limit value should be than the current limit for construction. The respondents were informed that a 10–20 per cent lower limit value would entail choosing either a low-carbon energy system or low-carbon building material, and that a 30–40 per cent lower limit value would necessitate low carbon solutions for both. A 10–20 per cent reduction gained most support (42%), followed by the more ambitious 30–40 per cent reduction (27%). Only seven per cent of the respondents felt that the limit value should not be any lower than the current standard of construction. However, many respondents (22%) did not know what the limit value should be. Open-ended comments revealed that they felt that the current carbon footprint of normal construction is poorly defined and that this made answering the limit value question hard, despite an explanation being provided. One respondent argued that 'determining an unambiguous threshold is extremely difficult. It can work in residential buildings, but otherwise the building stock is very diverse and different technical, qualitative, aesthetic, and other requirements have a significant effect on the end result'.

Fig. 3 illustrates the respondents' definite yes-responses[5] to different statements on the effects of the 10–20 per cent and 30–40 per cent limit values. Most of them agreed that a limit value would encourage builders to choose low-carbon building materials and material producers to develop low-carbon alternatives, and that it would increase the production of renewable energy and reduce the energy emissions of buildings. Many also believed that a limit value would have other benefits, including improvements in the quality of construction and increases in construction sector employment and Finnish low-carbon innovations. Such benefits of the regulation were considered more likely in the case of a 30–40 per cent limit value. However, there appears to be a trade-off, as more respondents believed that construction costs would also increase with a higher limit value.

The number of respondents who selected the 'do not know' option was always higher for the 30–40 per cent limit value than for the 10–20 per cent limit value. Many respondents were unsure whether solutions for meeting the limit value were already available on the market. This was especially true for the higher 30–40 per cent limit value, for which only 24 per cent of the respondents felt certain that solutions were already commercially available, whereas over 40 per cent felt the same about the 10–20 per cent limit value. Another argument for the 10–20 per cent limit value, raised in the open-ended questions, was that starting with a lower limit value and gradually raising it would better take into account the limitations of some building types to reducing their carbon footprint.

3.1.4. Construction priorities and the effects of the regulation

In addition to the questions about specific policy instruments, more

[5] The variable scale was yes – perhaps – no – do not know

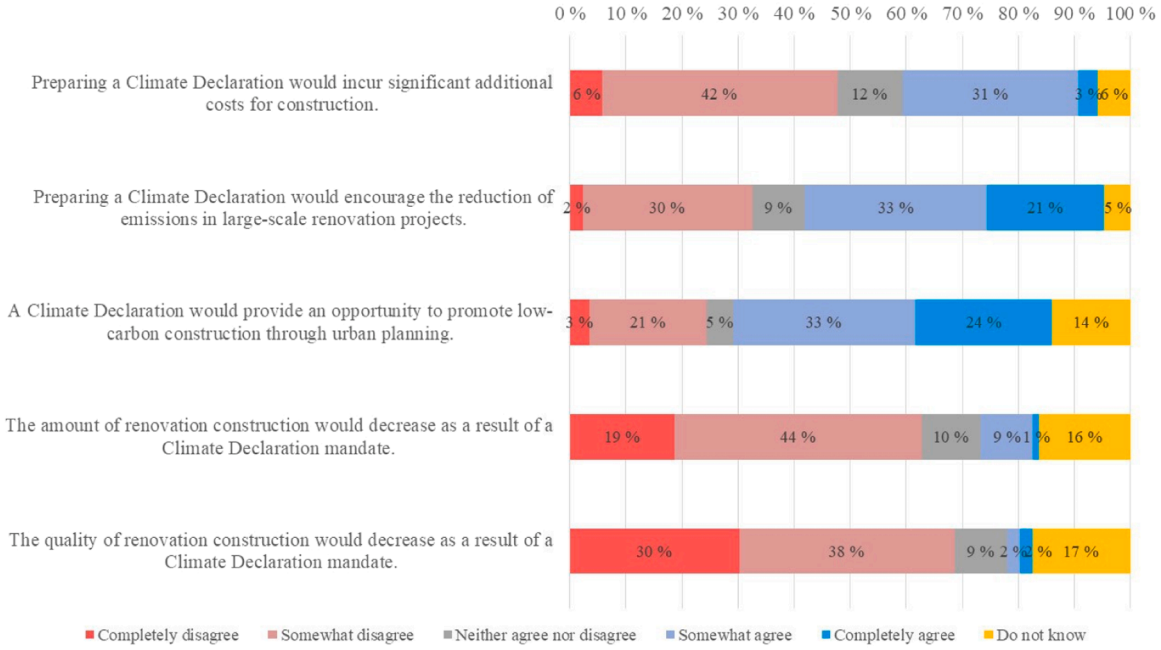


Fig. 2. Influences of Climate Declaration.

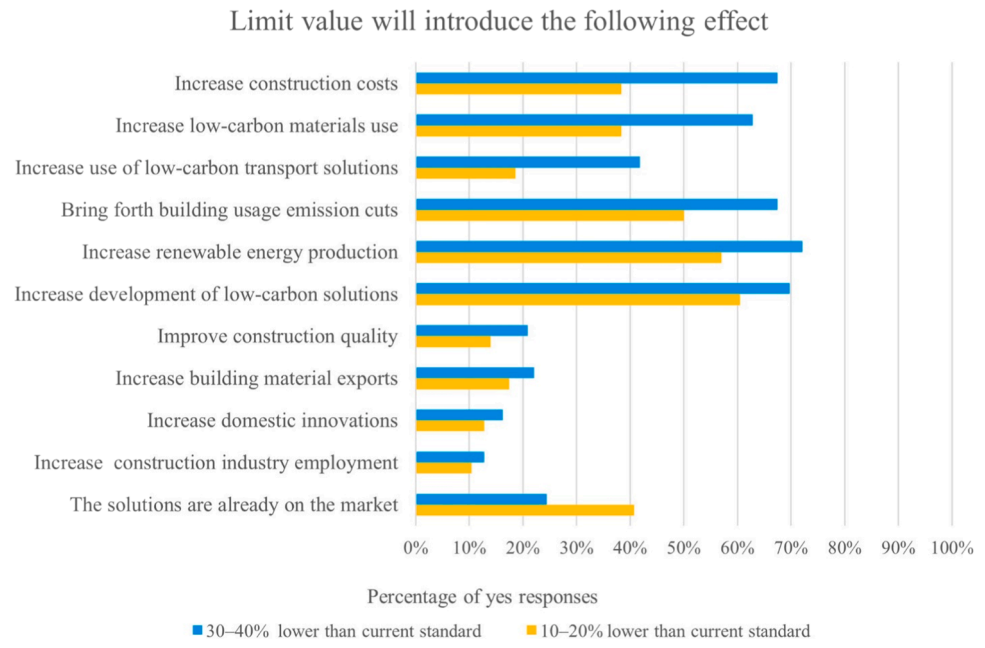


Fig. 3. Impacts of different limit value levels, from greatest difference to smallest.

general questions about the priorities of the respondents and their views on the new regulation in its entirety were also asked. Fig. 4 shows the results of the hypothetical question – if there was a choice between low emissions and other construction objective, which objective should be prioritised, or are the two objectives not in conflict with each other. The most critical issues prioritized over low emissions were indoor conditions and building longevity. As many as 58 per cent of respondents stated that indoor conditions should be prioritised, while 41 per cent did not consider the objectives to be in conflict with each other. In the open-ended questions, some of the respondents expressed concern about the quality of indoor conditions after energy renovations, especially if the ventilation system was changed. Building longevity was seen as a more important objective than low emissions by 53 per cent of the respondents, and 43 per cent saw no conflict between these targets. Many respondents raised the lifespan of the building as an important factor in both quality and life cycle emissions, but they also expressed concern that no data was yet available on the lifespan of the new construction solutions.

However, prioritizing another objective over the low-carbon objective does not necessarily mean that this will conflict with the proposed regulation.[6] The survey also asked the respondents what they believed

[6] This concern about the partial optimization and potential negative impacts on other aims of construction was evident in the public statements given during the two rounds of legal reform development.

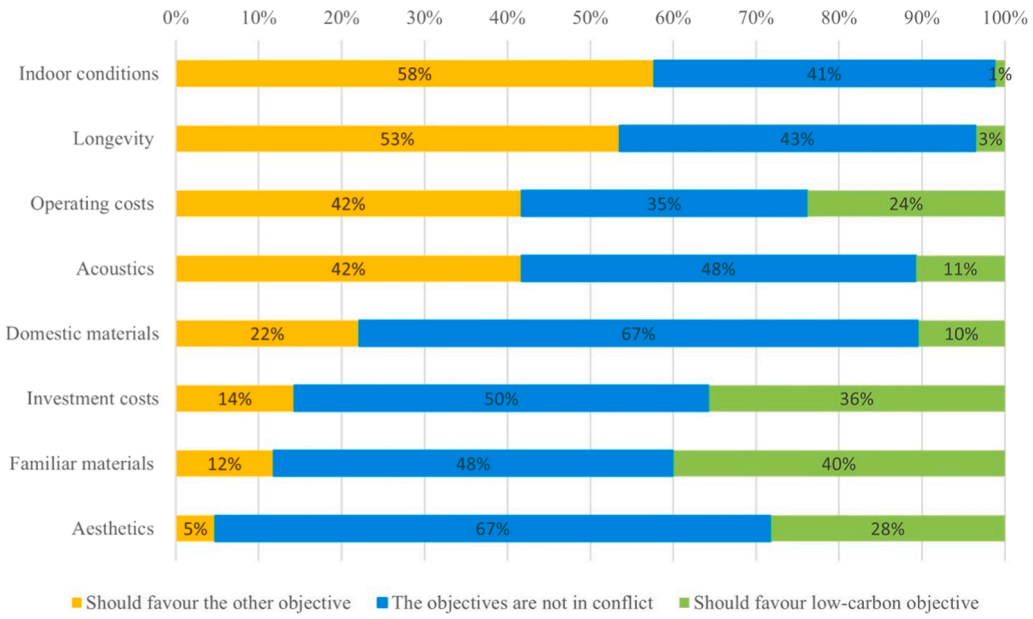


Fig. 4. Which objective of construction should be prioritized to avoid conflict; other-objective or low-carbon?.

to be the most important positive and negative effects of the new regulations (see Figs. 5 and 6).[7] Notably, only 6 (12%) of those who stated indoor conditions should be given preference if the objectives are in conflict selected suffering indoor conditions as one of the three most central negative impacts of the planned new regulations. Similarly, of the three central negative effects, operating costs, prioritised by 42 per cent of the respondents, were chosen by only three per cent of those respondents.

The most frequently chosen positive effects of the new regulation were the reduction of GHG emissions and the increased use of low-carbon energy solutions in buildings. These were chosen by over half of the respondents. Increased use of low carbon building materials was also frequently mentioned. Non-environmental benefits were generally not seen as key positive effects of the new regulations. For example, less than five per cent of the respondents chose increased employment and lower operating costs were also chosen by less than 5% of the respondents.

The responses to the question on the key negative effects of the regulation revealed clear concern about the state of current know-how and resources. Forty-five per cent of respondents believed that the authorities would be unable to monitor the changes, and 40 per cent believed that current expertise was not enough for the regulations. These findings are consistent with earlier research, which has identified insufficient knowledge and skills as key barriers to professionals' choice of low-carbon construction materials (Giesekam et al., 2016), zero-carbon homebuilding (Heffernan et al., 2015) and sustainable building (Häkkinen and Belloni, 2011). An assessment report by a Finnish building inspectors' professional organization warned that especially in small municipalities, building inspectors lack the knowledge and capacity to inspect the calculations of buildings' carbon footprints (RTY, 2017).[8]

3.2. The variation of the perceptions

To gain a deeper understanding of the overall perceptions of the new regulations, we focused on two key general statements:

i. *Fpsuitable*: The carbon footprint assessment method is an appropriate tool for assessing the climate impacts of construction.
ii. *Limdecemis:* A carbon footprint limit value is an effective way to reduce buildings' greenhouse gas emissions.

The sample mean scores on scale of 1–5, excluding the missing and do-not-know answers,[9] were 3.53 for *Fpsuitable* and 3.5 for *Limdecemis*. Thus, on average the respondents viewed the carbon footprint method as a suitable tool for assessing the impacts of building climate and the limit value as a functional method for reducing carbon emissions. However, this conclusion ignores the substantial variation in the responses, for which we conducted discrete choice regressions to study these differences.

The results imply that the differences did not stem from whether an actor was national or regional, working for a firm, or in the position of respondent in the organization. Organization size had a small positive impact on both, which is consistent with the findings of Chang et al. (2018) that in China, larger organizations view similar regulation more positively. Regarding construction material and type of construction in which the respondent's organization was involved, the immediate business impact of the regulation seemed to somewhat influence the perceptions, i.e., positive regression coefficient for operating in renovation and wooden construction and negative coefficient for block-house sector involvement. However, our results were heavily driven by respondent's organization type, which explains most of the variation in choice (see Appendix B). Table 3 lists the responses across the organization types.

A closer look at Table 3 reveals that organization types had similar mean scores, depending on where they were situated in the value chain of construction.[10] This warrants analysing them together. In the following, we examine consultants, architects, and other building design

[7] In total, a similar number of positives and negatives were selected (mean: 2,76 and 2,72).

[8] Similar concern was raised in expert interviews that we conducted as a part of the climate impact assessment. Five experts were asked how they saw the role of building inspection in the new carbon footprint regulation, and all five expressed concern – ranging from mild to very strong – about the lack of time and knowledge resources of the municipal building inspection authorities.

[9] Excluded across all formulations below.

[10] This grouping phenomenon is reflected throughout the questionnaire and is not limited to these variables. We will discuss these additional differences later.

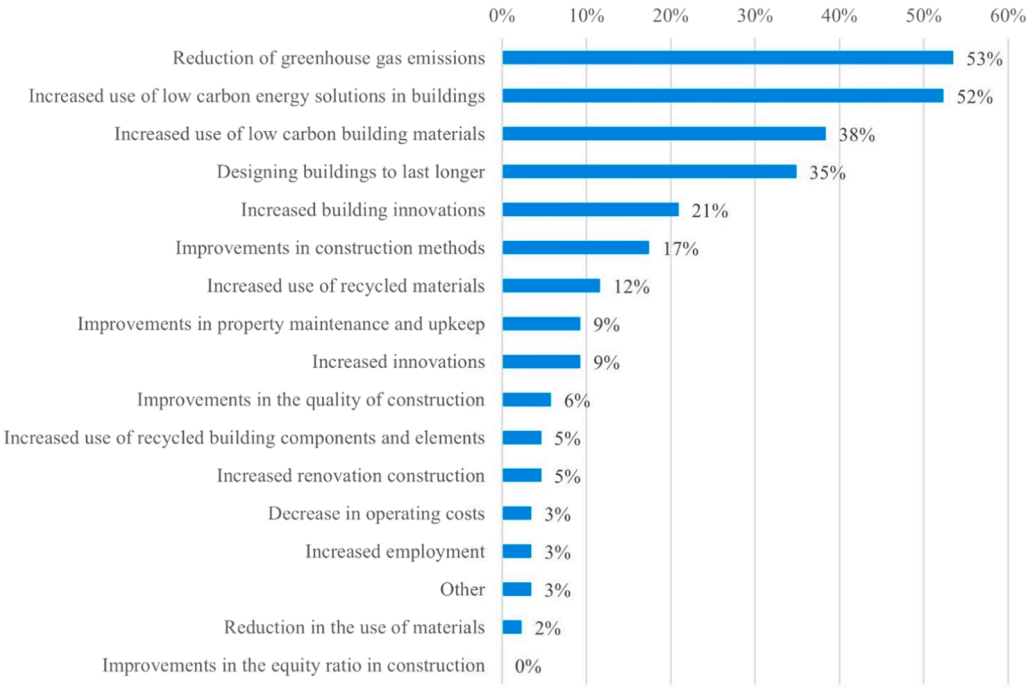


Fig. 5. What are the key positive effects of the new regulations? Choose max. three options.

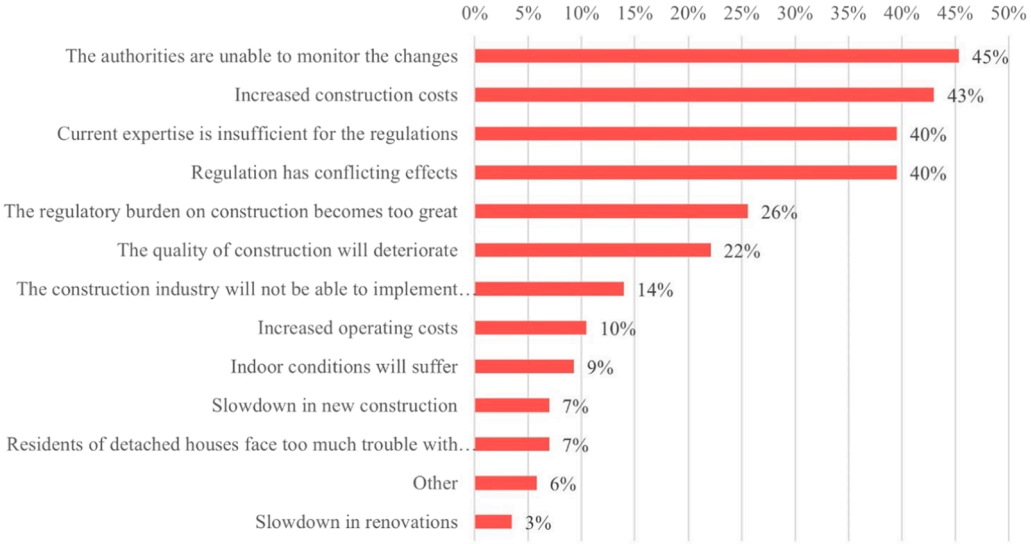


Fig. 6. What are the key negative effects of the new regulations? Choose max. three options.

companies jointly via the *Designers* variable. Companies directly involved in construction and material production form the *Construction* variable. This is consistent with Xu et al. (2019), in whose study the construction stage stakeholders were contractors and material suppliers. The associations and advocacy groups – *Advocates* – were examined separately. Finally, the remaining organizations were either listed under the *Others* category or examined individually.[11]

In the responses, the differences between the two variables within organization types were not great. However, the differences across organization types were extremely evident. Table 5 formalizes this by listing the statistical significance of the differences in the organizational views of the *Designers*, *Construction,* and the *Advocates*. Statistically significant differences could be detected across the board for the pairwise comparison assuming independent samples as measured using mean-rank Mann–Whitney test score analysis.

Generally, the *Designers* appeared to be more positive about the proposed regulations, *Construction* followed the sample means more closely, and the *Advocates* were more cautious. When we examined other variables, consistent with the *Limdecemis* and *Fpsuitable* views, we found that the *Advocates* tended to see fewer positive impacts of 10–20 per cent or 30–40 per cent limit values than the rest of the sample, whereas the *Designers* saw more benefits. The *Advocates* had no answer yet to how much lower the limit value should be than the current construction type.

The views on whether a Climate Declaration would encourage

[11] One could argue that research institutes could also belong to *Designers*. Their views in this space seemed to align more often with those of this group. However, for them, these policies are not so clearly a part of business development, which supports their separate examination.

Table 3
Method suitability across organization types.

Organization	*Limdecemis*	*Fpsuitable*
Construction company	3.00 (1.12, 9)	3.33 (0.87, 9)
Real estate	3.75 (1.26, 4)	3.67 (1.53, 3)
Readyhouse	5.00 (0.00, 1)	5.00 (0.00, 1)
Construction product industry	3.29 (1.42, 21)	3.50 (1.32, 20)
Architect agencies	4.67 (0.58, 3)	5.00 (0.00, 2)
Other design offices	4.71 (0,49, 7)	4.00 (1.00, 7)
Rental houses	4.00 (0.00, 1)	4.00 (0.00, 1)
Building supervision	2.00 (0.00, 2)	3.00 (0.00, 1)
Other government authority	1.00 (0.00, 2)	5,00 (0.00, 1)
Research institutes	4.14 (1.07, 7)	4.33 (1.21, 6)
Consulting firms	4.36 (0.50, 11)	4.36 (0.50, 11)
Association or advocacy	2.37 (1.12, 11)	2.24 (0.97, 17)
Others	2.50 (2.12, 2)	4.00 (0.00, 2)

*in brackets standard error, N

Table 4
Method suitability with pooled organization types.

Variable	Mean	Median	Mode	Standard dev.
Limdecemis				
Sample (N = 80)	3.5	4.00	2	1.341
Designers (N = 21)	4.52	5.00	5	0.512
Construction (N = 32)	3.19	3.00	2	1.378
Advocates (N = 11)	2.37	2.00	2	1.12
Researchers (N = 7)	4.14	4.00	5	1.069
Others (N = 9)	3.11	4.00	4	1.364
Fpsuitable				
Sample (N = 81)	3.53	4.00	4	1.256
Designers (N = 20)	4.30	4.00	4	0.733
Construction (N = 31)	3.52	4.00	4	1.391
Advocates (N = 17)	2.24	2.00	2	0.970
Researchers (N = 6)	4.33	5.00	5	1.211
Others (N = 7)	3.86	4.00	4	1.069

emission cuts in major renovations were also remarkably divided (Sample means: *Designers* 4,14; *Construction* 3,41 and *Advocates* 2,53). The *Advocates* also tended to prefer a more limited regulatory scope, and many wanted to exclude recreational buildings and detached houses from the planned regulations.

Other notable differences were that nearly 90 per cent of the *Designers* saw lower carbon emissions as one of the three most important positive impacts of the regulation, whereas the corresponding rate among *Construction* and research institutions was only about 30 per cent. Similarly, as regards low carbon building materials or low-carbon energy system solutions, *Construction* saw these as positive impacts much less often than the *Designers*. *Construction* and the *Advocates* also saw long building lifespan as a positive outcome of the planned regulations far more often than the *Designers*. *Construction* and the *Advocates* found three central positive impacts less often. Interestingly, however, the *Advocates* did not see construction cost increases as negative impacts as often as the other organizations.

However, there was less discrepancy among the views on the negative impacts of the regulations. One notable exception was that the *Advocates* and construction product industry firms were the only ones to raise the issue of reduction in construction quality as a potential negative outcome. Another major difference was that *Advocates* nearly uniformly (94%) predicted that the new regulations would have conflicting impacts, whereas considerably fewer respondents (27%) from the other organizations raised this point.

Although the results reported above offer important insights into climate policy perceptions, there are several limitations that should be kept in mind in their interpreration. First, although we took great care in designing and analysing the survey, as the survey mode directly targeted construction stakeholders, and due to the general limitations of using a survey as a research method, response or non-response bias may have arisen. Furthermore, although the number of responses from different organizations and the response rate implied fairly good representation, any conclusions on representativeness should be drawn with caution. This is due to the sheer impossibility of perfectly designating the sample population of construction sector stakeholders, or one person answering on behalf of the organization. Furthermore, as the survey was conducted in Finland, the results cannot necessarily be generalized directly to different geographies or cultures, or political, corporate, or legal systems and legal reforms.

4. Conclusions

This paper examines the perceptions of the actors in the construction sector on the emerging building carbon footprint regulation using survey data collected in Finland. We found that although on average, construction sector organizations viewed new carbon footprint policies positively, different organization types saw them very differently, and advocacy group that includes associations and interest groups generally saw them in a negative light. These differences may influence both the acceptance and the development of the planned regulations, especially as advocacy groups are powerful actors in Finnish climate policy (Vesa et al., 2020).

It is important to understand the views 'in the field' when new policies are planned. Finland has previously received praise for multi-stakeholder involvement in policy design. However, more research is needed on the impacts of stakeholder involvement in the final policies. For instance, legal development is at the very core of advocacy organizations, and they are heavily represented in the public statements on policy development. It is possible that their more cautious approach slows down critical policy implementation in a climate crisis that necessitates urgent action. Also in the forest sector, advocacy organizations have been more cautious in their environmental perceptions than the industries they represent (Kangas et al., 2018; Cashore and Vertinsky, 2000). Thus, this issue is not limited to the construction sector. Further research should examine whether this phenomenon holds in different sectors and geographies, and whether it persists further into legal development and all the way to policy implementation. Further research is also warranted on the relative influence of different stakeholder groups on the final legal product.

Another interesting finding of this paper was the fact that the construction sector actors who took part in the planning and design part of construction had a very positive outlook on the carbon footprint policies. This is important, because in a construction project, the main burden of the new policies falls on the planning and design part of the project. Thus, the willingness of construction sector actors to accept the new policies could help smooth the transition when the new policies are implemented in real-life construction projects. On the other hand, the

Table 5
Statistical significance of pairwise differences in views on method suitability and tool appropriateness.

	Designers		***Construction***		***Advocates***	
	Limdecemis	*Fpsuitable*	*Limdecemis*	*Fpsuitable*	*Limdecemis*	*Fpsuitable*
Designers	NA	NA	* **	* *	* **	* **
Construction	* **	* *	NA	NA	*	* **
Advocates	* **	* **	*	* **	NA	NA

* , * * and * ** represent statistically significant differences in views on 10%, 5% and 1% levels, respectively

construction sector actors in our study raised concerns about the authorities' competence in monitoring the planned regulations. Thus, proper resources should be designated to the parties responsible for implementing the monitoring.

Surprisingly, the most binding planned policy tool – the carbon footprint limit value – was largely approved by the construction sector actors in our study. On average, only the advocates did not perceive it as an effective policy instrument. For detached houses and recreational buildings, most of the construction sector actors did the not see the need for binding carbon footprint limit values. However, voluntary policies may result in low implementation rates (Haapio and Viitaniemi, 2008; Díaz-López et al., 2021). In a previous assessment of building carbon footprint regulation in Finland, Häkkinen and Vares (2018) concluded that non-binding carbon footprint policies are likely to have very small effects on building sector emissions, although they may be valuable transition tools for piloting and learning. This potential transitory benefit for voluntary climate policies is generally recognized (Thalmann and Baranzini, 2004). Significant emission cuts are only acquired when the policy widely affects the building sector, and a large share of constructors have to considerably adjust their operations.

The construction sector actors had concerns about the carbon footprint regulations overlapping with energy efficiency regulations. These concerns warrant further evaluations. Any overlapping regulation arising from, for example, future emissions trading mechanisms in the heating of buildings could also be relevant. One could argue that a more holistic regulation, encompassing the whole life cycle encourages the creation of solutions in the stages and segments in which they are most efficient. These may also more readily encompass the circular economy. On the one hand, we need to make sure that the sustainability transition speeds up. On the other hand, if it can be established that one regulation mechanism is sufficient to cut emissions, further regulatory burden should be avoided.

CRediT authorship contribution statement

Jouni Räihä: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Selina Clarke**: Formal analysis, Investigation, Visualization, Writing – review & editing. **Paula Sankelo:** Conceptualization, Investigation, Writing – review & editing. **Enni Ruokamo**: Conceptualization, Investigation, Writing – review & editing. **Hanna-Liisa Kangas:** Conceptualization, Investigation, Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Targeted respondents

Table A.1
Targeted respondents.

Targeted respondents	Reference
List of member organizations of the Confederation of Finnish Construction Industries (RT)*	https://www.rakennusteollisuus.fi/Rakennusteollisuus-RT/jasenyrityshaku/
List of largest construction companies in Finland*	https://www.rakennuslehti.fi/2018/07/suomen-kiinnostavimmat-rakennusliikkeet-ovat-yit-lehto-group-ja-srv/
List of largest detached house suppliers in Finland*	https://www.suomirakentaa.fi/talopaketit/talopaketin-valinta/suurimmat-talotoimittajat
List of largest architecture firms in Finland*	https://www.kauppalehti.fi/uutiset/arkkitehtitoimistot-porskuttavat-liiketoiminta-erinomaisen-kannattavaa/1b14bdae-c540–3d32–97ac-f704058c0c30
Organizations that took part in Level(s) testing in Finland*	https://julkaisut.valtioneuvosto.fi/handle/10024/161783
Public statements given on the Finnish carbon footprint calculation method	https://www.lausuntopalvelu.fi/FI/Proposal/Participation?proposalId= 8631c68f-8443–42c2–8b7c-b34ade99a3b2&proposalLanguage=da4408c3–39e4–4f5a-84db-84481bafc744
Emissions database for construction: collaboration organizations	https://www.co2data.fi/ and Finnish Environment Institute's collaborators' contact details
Overview of industrial wood construction – wood elements*	https://julkaisut.valtioneuvosto.fi/handle/10024/162338
Search engine (Google) to find other designers	Search terms [indicative translations in brackets]: 'energiasuunnittelu rakennukset'[energy design buildings] , 'rakennussuunnittelu' [building engineering], 'hiilijalanjälkilaskenta rakennukset [carbon footprint measurement buildings]'

*These were used to find the relevant organizations. The contact details of the respondents were found on their organizations' websites.

Appendix B. Binary Probit Models on determinants of positive view on building life cycle carbon footprint calculation

Table B.1
Binary Probit models of life cycle carbon footprint view.

	Limpositive (*Limdecemis* = 4 or 5 on 5-point Likert scale)		FPpositive (*FPsuitable* = 4 or 5 on 5-point Likert scale)	
Explanatory variable	Marginal effect	Standard error	Marginal effect	Standard error
Operating in renovation sector (1 if yes)	0.03777	0.09338	0.08538	0.09740
Work is related to wooden construction (1 if yes)	0.10621	0.10061	0.02012	0.09889
Operating in block-house sector (1 if yes)	-0.21111 * *	0.10193	-0.22602 * *	0.10116

(*continued on next page*)

Table B.1 (*continued*)

	Limpositive (*Limdecemis* = 4 or 5 on 5-point Likert scale)		FPpositive (*FPsuitable* = 4 or 5 on 5-point Likert scale)	
Operating all over Finland (1 if yes)	-0.08213	0.10172	0.00294	0.09460
Respondent works in upper management (1 if yes)	0.01897	0.09685	0.05131	0.08820
Organization employees (4 levels, <10, 10–50, 51–250 or <250)	0.04897 *	0.02681	0.05304 *	0.03010
Working in a firm (1 if yes)	-0.08917	0.10524	-0.08037	0.09661
Designers (1 if yes)	0.54662 * **	0.06452	0.40961 * **	0.10113
Advocates (1 if yes)	-0.28207 * **	0.10318	-0.49373 * **	0.13559
Observations (N)	79		80	
McFadden pseudo r2	0.35		0.35	
Correct predictions at 0.5	70%		71%	
Restricted log-likelihood	-52		-53	

Notes:
[a] Statistical significance at the 1%= ** *, 5%= ** or 10%= * risk level.
[b] Average Marginal effects, which denote change in the predicted probability of choosing the dependent variable, with the all the other variables fixed at the means when the independent variable changes from the baseline.
[c] Do-not-know responses and missing responses were excluded across all formulations.

References

Ankrah, N.A., Langford, D.A., 2005. Architects and contractors: a comparative study of organizational cultures. Constr. Manag. Econ. 23 (6), 595–607. https://doi.org/10.1080/01446190500126973.

Azari, R., Abbasabadi, N., 2018. Embodied energy of buildings: a review of data, methods, challenges, and research trends. Energy Build. 168, 225–235. https://doi.org/10.1016/j.enbuild.2018.03.003.

Balane, M.A., Palafox, B., Palileo-Villanueva, L.M., McKee, M., Balabanova, D., 2020. Enhancing the use of stakeholder analysis for policy implementation research: Towards a novel framing and operationalised measures. BMJ Glob. Health 5 (11). https://doi.org/10.1136/bmjgh-2020-002661.

Bijlsma, R., Bots, P., Wolters, H., Hoekstra, A., 2011. An empirical analysis of stakeholders' influence on policy development: the role of uncertainty handling. Ecol. Soc. 16 (1), 51. https://doi.org/10.5751/ES-03865-160151.

BPIE (Buildings Performance Institute Europe), 2021b. Whole-life carbon: challenges and solutions for highly efficient and climate-neutral buildings. ⟨https://www.bpie.eu/publication/whole-life-carbon-challenges-and-solutions-for-highly-efficient-and-climate-neutral-buildings/⟩.

Cashore, B., Vertinsky, I., 2000. Policy networks and firm behaviours: Governance systems and firm reponses to external demands for sustainable forest management. Policy Sci. 33 (1), 1–30.

Chang, R., Zuo, J., Zhao, Z., Soebarto, V., Lu, Y., Zillante, G., Gan, X., 2018. Sustainability attitude and performance of construction enterprises: a China study. J. Clean. Prod. 172, 1440–1451. https://doi.org/10.1016/j.jclepro.2017.10.277.

D'Agostino, D., Mazzarella, L., 2019. What is a nearly zero energy building? Overview, implementation and comparison of definitions. J. Build. Eng. 21, 200–212. https://doi.org/10.1016/j.jobe.2018.10.019.

Díaz-López, C., Carpio, M., Martín-Morales, M., Zamorano, M., 2021. Defining strategies to adopt Level(s) for bringing buildings into the circular economy. A case study of Spain. J. Clean. Prod. 287, 125048 https://doi.org/10.1016/j.jclepro.2020.125048.

, 2018European Union, 2018. Directive (EU) 2018/844 of the European Parliament and of the Council of 30 May 2018 amending Directive 2010/31/EU on the energy performance of buildings and Directive 2012/27/EU on energy efficiency. Official Journal of the European Union (2018), L 156/75.

, 2010European Union, 2010. Directive 2010/31/EU of the European Parliament and of Council of 19 May 2010 on the energy performance of building (EPBD recast). Official Journal of the European Union (2010), L 153/13. https://eur-lex.europa.eu/eli/dir/2010/31/2021-01-01.

BPIE (Buildings Performance Institute Europe), 2021a. Nearly Zero: A review of EU Member State implementation of new build requirements. Available at: ⟨https://www.bpie.eu/publication/nearly-zero-a-review-of-eu-member-state-implementation-of-new-build-requirements/⟩.

European Commission, 2020. Level(s) Specimen Specialised Article: For Public Authorities and Decision-Makers. 1.6.2020. https://ec.europa.eu/environment/publications/levels-specimen-specialised-article-public-authorities-and-decision-makers_en.

European Commission, 2021. Level(s) – European framework for sustainable buildings. 28.6.2021. https://ec.europa.eu/environment/publications/brief-introduction-levels_en.

Fraser, C., Zarkada-Fraser, A., 2003. Investigating the effectiveness of managers through an analysis of stakeholder perceptions. J. Manag. Dev. 22 (9), 762–783. https://doi.org/10.1108/02621710310495766.

Garcia, J.F., Kranzl, L., 2018. Ambition levels of nearly zero energy buildings (nZEB) definitions: an approach for cross-country comparison. Buildings 8 (143). https://doi.org/10.3390/buildings8100143.

Giesekam, J., Barrett, J.R., Taylor, P., 2016. Construction sector views on low carbon building materials. Build. Res. Inf. Int. J. Res. Dev. Demonstr. 44 (4), 423–444. https://doi.org/10.1080/09613218.2016.1086872.

Greene, W.H. k, 2012 (international edition). Econometric analysis, Seventh ed.,. Prentice Hall/Pearson.

Haapio, A., Viitaniemi, P., 2008. A critical review of building environmental assessment tools. Environ. Impact Assess. Rev. 28, 469e482. https://doi.org/10.1016/J.EIAR.2008.01.002.

Habert, G., Röck, M., Steininger, K., Lupisek, A., Birgisdottir, H., Desing, H., Chandrakumar, C., Pittau, F., Passer, A., Rovers, R., Slavkovic, K., Hollberg, A., Hoxha, E., Jusselme, T., Nault, E., Allacker, K., Lützkendorf, T., 2020. Carbon budgets for buildings: harmonising temporal, spatial and sectoral dimensions. Build. Cities 1 (1), 429–452. https://doi.org/10.5334/bc.47.

Häkkinen, T., Belloni, K., 2011. Barriers and drivers for sustainable building. Build. Res. Inf. Int. J. Res. Dev. Demonstr. 39 (3), 239–255. https://doi.org/10.1080/09613218.2011.561948.

Häkkinen, T., Vares, S., 2018. *Rakennusten khk-päästöjen ohjauksen vaikutusten arviointi.* Impact assessment of different measures for controlling GHG emissions of buildings VTT Technical Research Centre of Finland. VTT Technol. (No. 324) ⟨http://www.vtt.fi/inf/pdf/technology/2018/T324.pdf⟩.

Heffernan, E., Pan, W., Liang, X., de Wilde, P., 2015. Zero carbon homes: perceptions from the UK construction industry. Energy Policy 79, 23–36. https://doi.org/10.1016/j.enpol.2015.01.005.

Jin, X., Zhang, G., Liu, J., Feng, Y., Zuo, J., 2017. Major participants in the construction industry and their approaches to risks: a theoretical framework. Procedia Eng. 182, 314–320. https://doi.org/10.1016/j.proeng.2017.03.100.

Kangas, H.-L., Kivimaa, P., Lazarevic, D., 2018. Technical skills, disinterest and non-functional regulation: barriers to building energy efficiency in Finland viewed by energy service companies. Energy Policy 114, 63–76. https://doi.org/10.1016/j.enpol.2017.11.060.

Koezjakov, A., Urge-Vorsatz, D., Crijns-Graus, W., van den Broek, M., 2018. The relationship between operational energy demand and embodied energy in Dutch residential buildings. Energy Build. 165, 233–245. https://doi.org/10.1016/j.enbuild.2018.01.036.

Kuittinen, M., Häkkinen, T., 2020. Reduced carbon footprints of buildings: new Finnish standards and assessments. Build. Cities 1 (1), 182–197. https://doi.org/10.5334/bc.30.

Kuittinen, M. (ed.), 2019. Method for the whole life carbon assessment of buildings. 20.9.2019. ⟨http://urn.fi/URN:ISBN:978–952-361–030-9⟩.

Lai, I.K., Lam, F.K., 2010. Perception of various performance criteria by stakeholders in the construction sector in Hong Kong. Constr. Manag. Econ. 28 (4), 377–391. https://doi.org/10.1080/01446190903521515.

von Malmborg, F., Rohdin, P., Wihlborg, E., 2023. Climate declarations for buildings as a new policy instrument in Sweden: a multiple streams perspective. Build. Res. Inf. Int. J. Res. Dev. Demonstr. 1–18. https://doi.org/10.1080/09613218.2023.2222320.

Ministry of the Environment (2021). Carbon Footprint Limits for Common Building Types. 11.1.2021. https://mrluudistus.fi/wp-content/uploads/2021/01/Bionova_MinEnv_Finland_embodied_carbon_limit_values_report_FINAL_19JAN2021_ed.pdf.

Olander, S., Landin, A., 2005. Evaluation of stakeholder influence in the implementation of construction projects. Int. J. Proj. Manag. 23 (4), 321–328. https://doi.org/10.1016/j.ijproman.2005.02.002.

Rasmussen, A., Toshkov, D., 2013. The effect of stakeholder involvement on legislative duration: consultation of external actors and legislative duration in the European Union. Eur. Union Polit. 14 (3), 366–387. https://doi.org/10.1177/1465116513489777.

Röck, M., Saade, M.R.M., Balouktsi, M., Rasmussen, F.N., Birgisdottir, H., Frischknecht, R., Habert, G., Lützkendorf, T., Passer, A., 2020. Embodied GHG emissions of buildings — the hidden challenge for effective climate change mitigation. Appl. Energy 258, 114107. https://doi.org/10.1016/j.apenergy.2019.114107.

RTY, 2017. Viranomaisnäkökulma rakennuksen elinkaaren hiilijalanjälkiohjaukseen. [Public authority view on regulating life cycle carbon footprint of a building] Rakennustarkastusyhdistys RTY ry. 29.6.2017. ⟨https://www.ymparisto.fi/download/noname/%7B0AF5142F-F2BD-4DD3–81CD-D341B31741A1%7D/129192⟩.

Serpell, A., Kort, J., Vera, S., 2013. Awareness, actions, drivers and barriers of sustainable construction in Chile. Technol. Econ. Dev. Econ. 19 (2), 272–288. https://doi.org/10.3846/20294913.2013.798597.

Thalmann & Baranzini 2004: Voluntary Approaches in Climate Policy (book chapter).

Train, K.E., 2009. Discrete Choice Methods with Simulation, Second ed.,. Cambridge University Press, p. 2009. https://doi.org/10.1017/CBO9780511805271.

United Nations Environment Programme 2021. 2021 Global Status Report for Buildings and Construction: Towards a Zero-emission, Efficient and Resilient Buildings and Construction Sector. Nairobi. ⟨https://globalabc.org/sites/default/files/2021–10/GABC_Buildings-GSR-2021_BOOK.pdf⟩.

Vesa, J., Gronow, A., Ylä-Anttila, T., 2020. The quiet opposition: how the pro-economy lobby influences climate policy. Glob. Environ. Change 63, 102117. https://doi.org/10.1016/j.gloenvcha.2020.102117.

Xu, X., Li, C.Z., Wang, J., Huang, W., 2019. Collaboration between designers and contractors to improve building energy performance. J. Clean. Prod. 219, 20–32. https://doi.org/10.1016/j.jclepro.2019.02.036.

Zhao, D., McCoy, A.P., Kleiner, B.M., Mills, T.H., Lingard, H., 2016. Stakeholder perceptions of risk in construction. Saf. Sci. 82, 111–119. https://doi.org/10.1016/j.ssci.2015.09.002.